

THE CYBER KILL CHAIN

FOR SOC ANALYSTS

WHAT IS THE CYBER KILL CHAIN?



The Cyber Kill Chain is a model that shows the steps an attacker follows to attack a target.

It has 7 stages from start to end.

It helps SOC analysts know what to look for and stop attacks early.

WHY IS IT IMPORTANT?

- ✓ It shows how attacks happen step by step.
- ✓ It helps us find weak points.
- ✓ It helps us detect attacks early.
- ✓ The earlier we stop, the less damage.



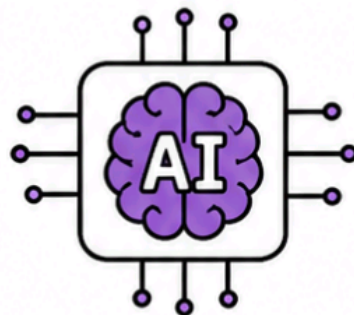
HOW IS IT USED IN SOC?



- ➔ We know what attackers are doing at each step.
- ➔ We set alerts and rules for each step.
- ➔ We investigate and stop the attack in early steps.
- ➔ We reduce damage and protect the organization.

HOW AI WILL HELP?

- ★ AI can analyze lots of data very fast.
- ★ AI can find weird or suspicious activities.
- ★ AI can predict attacks before they happen.
- ★ AI helps reduce false alerts.
- ★ AI helps SOC analysts work smarter and faster.



WHAT IS CYBER KILL CHAIN?



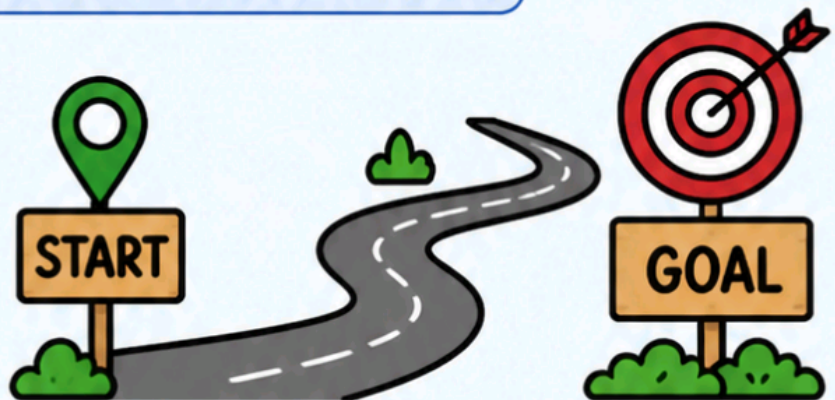
The Cyber Kill Chain is a **model** that shows the **steps** an **attacker** follows to **attack** a **target**.

SIMPLE EXPLANATION

It has **7** stages from start to end.
It helps **SOC analysts** know what to look for and **stop attacks early**.

THINK OF IT LIKE THIS

It is like a road map that shows the path an attacker takes from start to finish.



WHY IS IT IMPORTANT?



The Cyber Kill Chain is important because it helps us understand attacks step by step and stop them before they cause big damage.

IT IS IMPORTANT BECAUSE:



1. Understands Attacks

It shows how attackers think and act.



2. Finds Weak Points

It helps us find and fix weak areas in our systems.



3. Detects Early

It helps us detect attacks in early stages.



4. Reduces Damage

Early stopping means less damage and less loss.



5. Improves Security

It helps us build better security and stay safe.



The earlier we understand and stop, the safer we are!

HOW IS IT USED IN SOC?



In SOC (Security Operations Center), we use the Cyber Kill Chain to understand, detect and stop attacks in early steps.

IN SOC, WE USE IT LIKE THIS:

1



WE KNOW WHAT TO LOOK FOR

We know what attackers do in each step. So we can look for those activities in our systems.

2



WE SET ALERTS & RULES

We create alerts and rules for each step. If something suspicious happens, we get an alert.

3



WE INVESTIGATE FASTER

When an alert comes, we know which step it is. This helps us investigate faster and better.

4



WE STOP ATTACKS EARLY

If we stop the attack in early steps, we can prevent more damage.

5



WE IMPROVE & LEARN

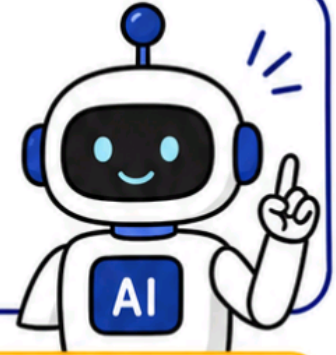
We learn from every attack and make our systems and rules stronger.



So, Cyber Kill Chain helps SOC analysts work **smarter**, **faster** and **more effectively** to protect the organization!



HOW AI HELPS IN CYBER KILL CHAIN?



AI (Artificial Intelligence) is like a super helper for SOC analysts. It looks at huge amounts of data quickly and helps us **detect**, **understand** and **stop** attacks in every step of the Cyber Kill Chain.

AI HELPS IN THESE WAYS:

1



FIND THREATS FASTER

AI looks at a lot of data in seconds and finds suspicious things that humans might miss.

2



SMART ALERTS

AI reduces fake alerts (false positives) and shows only the important alerts to analysts.

3



PREDICTS ATTACKS

AI can find patterns and predict possible attacks before they happen.

4



AUTOMATES REPETITIVE TASKS

AI does boring and time-consuming work like sorting logs, so analysts can focus on real threats.

5



IMPROVES DECISIONS

AI gives suggestions and insights, so analysts can take better and faster action.



In short, AI is like a **smart assistant** that helps SOC analysts work **faster**, **smarter** and keep our organization **safe**!



1

STAGE 1: RECONNAISSANCE



WHAT IS RECONNAISSANCE?

Reconnaissance is the first step of the Cyber Kill Chain. In this step, the attacker **collects information** about the **target** before planning an attack.

WHAT DOES THE ATTACKER DO?

- ✓ Finds details about the company or organization.
- ✓ Looks for email addresses, employees, job roles.
- ✓ Finds out which technologies, websites and tools are used.
- ✓ Scans open ports and servers.
- ✓ Collects information from social media, websites, news, etc.



EXAMPLES OF INFORMATION THEY LOOK FOR

- Email addresses
- Employee names and positions
- Company websites and domains
- Technologies and software used
- Open ports and servers
- Social media and public info

HOW DOES SOC ANALYST USE THIS STEP?



- ★ Analysts monitor for unusual scanning, searching or data collection activity.
- ★ They watch for port scanning, unknown IPs and suspicious requests.
- ★ Early detection here helps stop the attacker in the next steps.



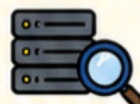
EXAMPLE



Attacker searches company website



Finds employees, emails, and tools



Scans open ports and systems



Collects all info for planning attack



Ready for next step!



Reconnaissance = Information is Power for the attacker.
Our Goal = **Detect** it early and **stop** it here!



2

STAGE 2: WEAPONIZATION



WHAT IS WEAPONIZATION?

Weaponization is the second step of the Cyber Kill Chain. In this step, the attacker creates a **malicious package** or **file** that will be used to attack the target.

WHAT DOES THE ATTACKER DO?

- ✓ Creates malicious files or programs like viruses, malware or scripts.
- ✓ Adds the malicious code to a normal-looking file or email.
- ✓ Makes the file look safe and real so people will trust it.
- ✓ Prepares the weapon (malware) to be delivered in the next step.



EXAMPLE



Attacker creates a fake invoice file. Inside that file, there is malware. When someone opens it, the malware starts working.



HOW DOES SOC ANALYST USE THIS STEP?

- ★ Analysts look for unusual or suspicious files.
- ★ They check email attachments and links before opening.
- ★ They use security tools to scan files and find hidden threats.
- ★ This helps stop the weapon before it reaches the target.



EXAMPLE FLOW



Adds malware inside the file



Makes it look normal and safe



Prepares it to send to target



Ready for next step (Delivery)



Weaponization = **Creating** the attack weapon.

Our Goal = **Detect** this weapon and **stop** it early!



3

STAGE 3: DELIVERY



WHAT IS DELIVERY?

Delivery is the third step of the Cyber Kill Chain.

In this step, the attacker sends the **malicious weapon** (file, link, or code) to the target.

The goal is to get the weapon into the target's system.

WHAT DOES THE ATTACKER DO?

- ✓ Chooses a way to send the malicious file to the target.
- ✓ Uses email, websites, USB drives, or other methods.
- ✓ Tricks the user to open or run the malicious file.
- ✓ Once opened, the attack moves to the next step (Exploitation).



EXAMPLES OF DELIVERY METHODS

- Phishing emails with malicious attachments or links
- Malicious websites or drive-by downloads
- Infected USB drives
- Instant messages or social media links
- Downloading files from untrusted sources



HOW DOES SOC ANALYST USE THIS STEP?

- ★ Analysts monitor email traffic, web traffic and file downloads.
- ★ They look for suspicious links, attachments and unusual activity.
- ★ They use tools to block malicious files and websites.
- ★ Early detection here stops the attack from reaching the system.



EXAMPLE FLOW



Sends it through email (phishing)



User receives the email



User opens the attachment or link



Weapon reaches the system



Delivery = Getting the weapon to the target.

Our Goal = **Detect** it and **stop** it before it is **opened**!



4

STAGE 4: EXPLOITATION



WHAT IS EXPLOITATION?

Exploitation is the fourth step of the Cyber Kill Chain.

In this step, the attacker uses the weapon (malicious file or code) to take advantage of a **weakness or vulnerability** in the target's system.

The goal is to get inside the system and gain control.

WHAT DOES THE ATTACKER DO?

- ✓ Finds a weakness (vulnerability) in the system or application.
- ✓ Uses the weapon to exploit that weakness.
- ✓ Runs malicious code on the target system.
- ✓ Gets access to the system (may be limited or full access).
- ✓ Tries to stay hidden and move to the next step (Installation).



EXAMPLES OF EXPLOITATION



Exploiting software bugs or vulnerabilities.



Using unpatched software or missing updates.



Exploiting weak passwords or misconfigurations.



Running malicious code or commands.



HOW DOES SOC ANALYST USE THIS STEP?

- ★ Analysts look for signs of exploitation attempts.
- ★ They monitor logs, alerts and unusual system behavior.
- ★ They check for suspicious commands, privilege escalation, or unexpected changes.
- ★ Early detection here can stop the attacker from gaining control.



EXAMPLE FLOW



Malicious file delivered



Exploits a vulnerability



Malicious code runs



Attacker gets access



Ready for next step (Installation)



Exploitation = Using the weakness to get inside the system.

Our Goal = **Detect** it and **stop** it before attacker gains control!



5

STAGE 5: INSTALLATION



WHAT IS INSTALLATION?

Installation is the fifth step of the Cyber Kill Chain. In this step, the attacker installs the **malicious program** (backdoor, remote access tool, or malware) inside the target's system. The goal is to make sure the attacker can stay inside the system for a long time.

WHAT DOES THE ATTACKER DO?

- ✓ Installs malware or a backdoor in the system.
- ✓ Creates a hiding place so the malware is hard to find.
- ✓ Starts the malware automatically when the system starts.
- ✓ Makes sure the attacker can come back anytime.
- ✓ Prepares the system for the next step (Command & Control).



EXAMPLES OF INSTALLATION



Installing a backdoor to come back later.



Changing system settings to run malware automatically.



Copying malware files to safe hidden locations.



Making malware start every time the system boots up.

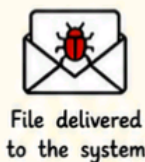


HOW DOES SOC ANALYST USE THIS STEP?

- ★ Analysts look for new or unusual programs and files.
- ★ They check if any unknown program starts automatically.
- ★ They monitor system changes and new user accounts.
- ★ They use security tools to detect hidden malware or backdoors.



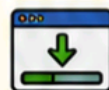
EXAMPLE FLOW



File delivered to the system



Attacker exploits a weakness



Malware or backdoor is installed



Malware hides in the system



Malware runs every time



Installation = Setting up a place to stay in the system.

Our Goal = Detect it and **stop** it before attacker comes back!



6 STAGE 6: COMMAND & CONTROL



WHAT IS COMMAND & CONTROL?

Command & Control (C2) is the sixth step of the Cyber Kill Chain. In this step, the attacker sets up a **communication channel** with the compromised system.

The goal is to control the system remotely and send commands to perform more actions.

WHAT DOES THE ATTACKER DO?

- ✓ Creates a connection between the attacker and the victim's system.
- ✓ Uses this connection to send commands to the infected system.
- ✓ Steals data, uploads files or performs other malicious actions.
- ✓ Keeps the connection active to maintain control.
- ✓ Tries to hide this communication from security tools.



EXAMPLES OF COMMAND & CONTROL

-  Connecting to a remote server controlled by the attacker
-  Using web traffic (HTTP/HTTPS) to send commands
-  Using encrypted channels to hide communication
-  Using cloud services to control the infected system
-  Using unusual ports or protocols to avoid detection



HOW DOES SOC ANALYST USE THIS STEP?

- ★ Analysts look for unusual outbound connections.
- ★ They check for unknown domains, IPs and ports.
- ★ They monitor for suspicious protocols and encrypted traffic.
- ★ They use threat intelligence to identify C2 servers.
- ★ Early detection here can stop the attacker from controlling the system.



EXAMPLE FLOW



Command & Control = Attacker **controls** the infected system remotely.
Our Goal = Detect the connection and **stop** the attacker!



7 STAGE 7: ACTIONS ON OBJECTIVES



WHAT ARE ACTIONS ON OBJECTIVES?

Actions on Objectives is the final and seventh step of the Cyber Kill Chain.

In this step, the attacker **achieves their goal** by using the access and control they have over the target's system.

The goal is to cause damage, steal data, or disrupt operations.

WHAT DOES THE ATTACKER DO?

- ✓ Steals sensitive data (files, passwords, personal info, etc.).
- ✓ Deletes or encrypts important files (ransomware).
- ✓ Changes data or system settings.
- ✓ Disrupts services or makes the system unavailable.
- ✓ Uses the system for other attacks (spreading malware, attacking others).
- ✓ Hides their tracks to avoid detection.



EXAMPLES OF ACTIONS ON OBJECTIVES



Stealing company data or customer information



Demanding money (ransomware attacks)



Deleting important files or databases



Disrupting websites, servers or business operations



Using the system to launch other attacks



Clearing logs and hiding evidence



HOW DOES SOC ANALYST USE THIS STEP?

- ★ Analysts look for unusual or harmful activities.
- ★ They check for data theft, file changes, or service disruptions.
- ★ They monitor alerts for ransomware or data exfiltration.
- ★ They investigate and respond quickly to reduce damage.
- ★ Early action here helps protect important data and systems.



Attacker has full control



Attacker steals or encrypts data



EXAMPLE FLOW



Attacker disrupts or damages system



Attacker uses system for other attacks



Attacker hides their tracks



Actions on Objectives = **Attacker achieves their final goal.**
Our Goal = **Detect the attack and stop the damage!**





SUMMARY

THE 7 STEPS OF CYBER KILL CHAIN



STEP	WHAT IT MEANS	WHAT THE ATTACKER DOES
1 RECONNAISSANCE 	The attacker learns about the target.	 Collects information about the victim (such as people, systems, websites).
2 WEAPONIZATION 	The attacker creates a weapon (malicious file or code).	 Creates a malicious file or link to harm the victim.
3 DELIVERY 	The attacker delivers the weapon to the victim.	Sends the malicious file or link through email, messages, websites, etc.
4 EXPLOITATION 	The attacker takes advantage of a weakness.	Uses the weapon to exploit a weakness in the system.
5 INSTALLATION 	The attacker installs malware inside the system.	Installs malicious program so the attacker can stay inside the system.
6 COMMAND & CONTROL (C2) 	The attacker connects to the infected system remotely.	Sets up communication with the system and sends commands.
7 ACTIONS ON OBJECTIVES 	The attacker achieves their goal.	Steals data, damages systems, or disrupts operations.



MAIN GOAL OF THE CYBER KILL CHAIN:

The attacker's goal is to gain access, stay inside, and achieve their objective without being detected.



HOW CAN WE PREVENT THESE ATTACKS?

- ✓ Keep software and systems updated.
- ✓ Use strong passwords.
- ✓ Be careful with emails and links.
- ✓ Use antivirus and firewall.
- ✓ Monitor systems and network.
- ✓ Report anything suspicious.



AWARENESS + PREVENTION = CYBER SECURITY

Stay Alert. Stay Safe. Stay Secure!



FINAL WORDS - YOU ARE NOW ONE STEP AHEAD!

★ Congratulations! You have completed your notes on **CYBER KILL CHAIN**. ★



Every expert was once a beginner.

Keep learning, keep practicing,
and soon you will be the analyst others look up to!



WHAT YOU LEARNED

- ✓ Cyber Kill Chain - 7 steps
- ✓ Phishing basics & how it works
- ✓ Email header analysis
- ✓ Link & URL investigation
- ✓ Attachment analysis
- ✓ User behavior & indicators
- ✓ Tools used by SOC analysts
- ✓ Real-world scenarios
- ✓ Verdict & action steps
- ✓ SOC workflow & best practices



WHY THIS KNOWLEDGE MATTERS

- ★ Attackers follow the Cyber Kill Chain to achieve their goal.
- ★ Understanding each step helps you **detect** and **stop** attacks early.
- ★ Good analysts protect users, data and organizations.
- ★ Your work can stop attackers before they cause real damage.



YOUR NEXT STEPS

- ① **PRACTICE DAILY** → Analyze real phishing emails, headers, links & files.
- ② **USE LABS** → Practice in safe labs to improve your skills.
- ③ **STAY UPDATED** → Learn new tools, techniques and attack trends.
- ④ **BUILD PORTFOLIO** → Document your investigations and learnings.
- ⑤ **BECOME JOB-READY** → Consistency today, Career tomorrow!



GOLDEN RULES (ALWAYS REMEMBER)

- ✓ Think before you click.
- ✓ Verify before you trust.
- ✓ Report to protect.
- ✓ Stay alert, stay safe.
- ✓ Knowledge + Practice = Mastery!



MINDSET OF A GREAT ANALYST

- ★ Be Curious
- ★ Be Detail Oriented
- ★ Be Patient
- ★ Be Logical
- ★ Be Passionate



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“Better you understand attacks,
better you can defend them.”

Keep Learning. Keep Growing. Keep **Defending!** 😊



Thank You for completing these notes! You are Awesome!

